

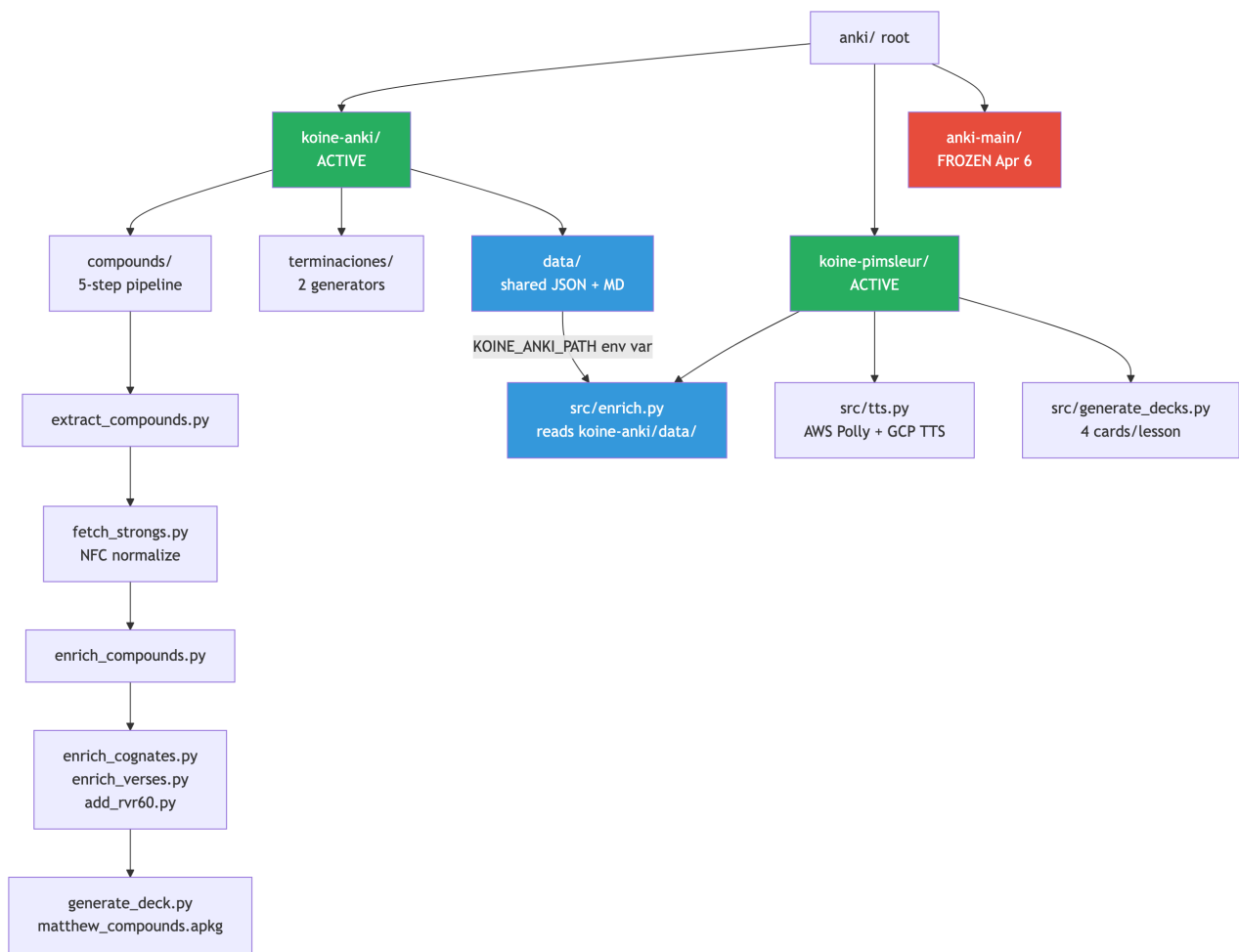
anki-agent Enhancement Report

Investigation ID: 7f55f0cd | Date: 2026-05-24

Executive Summary

The current `anki-agent` prompt (1,159 chars) contains **2 confirmed factual errors** and **4 significant gaps**. Four parallel agents investigated project structure, deck pipelines, TTS architecture, and card quality standards. The enhanced prompt (2,847 chars) corrects all errors and adds the missing context. The most critical fix: the prompt incorrectly references `bible-tools` and `Wavenet-B` — neither exists in the actual codebase. The active projects are `koine-anki/` and `koine-pimsleur/`; `anki-main/` is frozen as of April 6.

Architecture Overview



Detailed Findings

Project Structure

Subproject	Status	Purpose
<code>koine-anki/</code>	✅ Active	Compound word Anki decks + grammar endings
<code>koine-pimsleur/</code>	✅ Active	90-lesson Pimsleur audio course + Anki decks
<code>anki-main/languages/</code>	❌ Frozen (Apr 6)	Original monorepo — do not edit

`koine-anki/compounds/` is the active fork of the compounds pipeline. Five scripts are byte-identical with `anki-main/`; `generate_deck.py` has diverged — `koine-anki` is newer with cognates + verses sections.

Deck Generation Tools

Script	Output	Format
<code>koine-anki/compounds/</code> <code>generate_deck.py</code>	<code>decks/</code> <code>matthew_compounds.apkg</code>	.apkg (genanki)
<code>koine-anki/terminaciones/</code> <code>gen_deck.py</code>	<code>terminaciones/</code> <code>greek_infinitives_anki.txt</code>	TSV (text import)
<code>koine-anki/terminaciones/</code> <code>gen_deck_vol3.py</code>	<code>terminaciones/</code> <code>terminaciones_vol3.apkg</code>	.apkg (genanki)
<code>koine-pimsleur/src/</code> <code>generate_decks.py</code>	<code>decks/anki/*.apkg</code>	.apkg (genanki)

Pimsleur card selection per lesson: 2 vocab → 1 phrase → 1 verse → fill from remaining vocab. Both Recognition (Greek→Spanish) and Production (Spanish→Greek) templates generated.

Stable Deck IDs (never change these)

Deck	Model ID
Koiné Pimsleur	<code>1646410900000</code>
Greek Compounds + Cognates	<code>1646410861305</code>
Terminaciones Vol.3	<code>1607392319</code>

TTS Architecture

- **Spanish narration:** AWS Polly — voice `Mia`, neural, `es-MX`
- **Greek speech:** Google Cloud TTS — Chirp3-HD voices (`Achird` / `Zephyr` primary; `Charon` / `Aoede` / `Kore` fallback pool)

- **Short-word fallback trigger:** character count `< 4` Greek chars (NOT duration-based)
- **Speed progression:** 0.75x (L1-7) → 0.85x (L8-15) → 1.0x (L16+)
- **Bedrock model:** `us.anthropic.claude-sonnet-4-20250514-v1:0` (us-east-1, max_tokens=12000)
- **QA log:** `/tmp/tts_cache/tts_qa_report.txt`

Cross-Repo Dependency

`koine-pimsleur/src/enrich.py` reads from `koine-anki/data/`: - `koine-anki/data/matthew/compounds_enriched.json` - `koine-anki/data/morphology_reference.md`

Resolved via `KOINE_ANKI_PATH` env var (defaults to sibling directory placement). No `bible-tools` dependency exists.

Errors in Current Prompt

Error 1 — bible-tools Reference (WRONG)

Current prompt says: > “koine-anki/data/ is referenced by bible-tools ingest_local.py”

Reality: `koine-pimsleur/src/enrich.py` consumes `koine-anki/data/` via `KOINE_ANKI_PATH` env var. No `bible-tools` reference exists anywhere in this project.

Error 2 — Wavenet-B Fallback (WRONG)

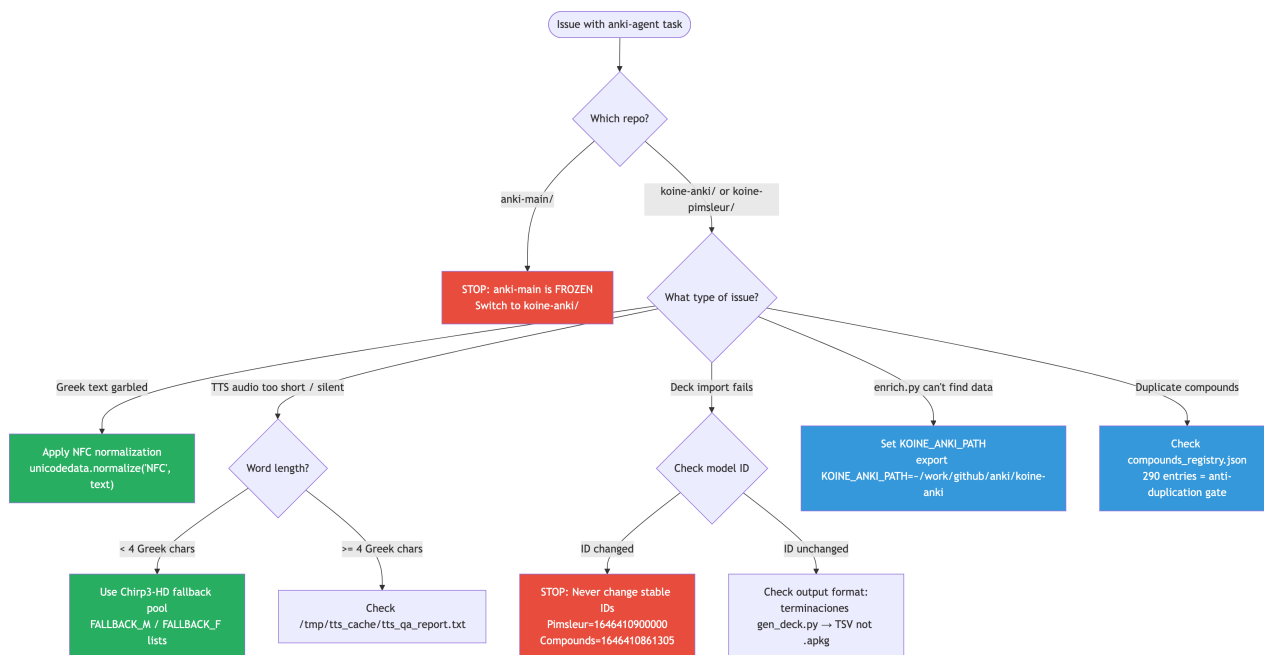
Current prompt / notes say: > “Chirp3-HD fails on short words → Wavenet-B fallback”

Reality (verified in `tts.py`): - Trigger: character count `< 4` Greek chars (not duration) - Fallback: other Chirp3-HD voices from `FALLBACK_M` / `FALLBACK_F` lists — NOT Wavenet-B - Duration formula: `max(400, min(n,4)*150 + max(0,n-4)*80)` ms

Gaps in Current Prompt

Gap	Impact	Fix
G1 — No <code>.kiro/steering.md</code>	Agent lacks always-on NFC + blacklist context	Create steering file
G2 — Broken venv shebang	<code>pip</code> fails in anki-main	Use <code>python3 -m pip</code>
G3 — 4 new koine-anki scripts not mentioned	Agent unaware of <code>enrich_cognates.py</code> , <code>audit_cognates.py</code> , <code>enrich_verses.py</code> , <code>add_rvr60.py</code>	Add to prompt
G4 — Lesson data format undocumented	Agent can't safely modify lesson dicts	Document in notes

Troubleshooting Decision Tree



Action Plan

Action 1 — Apply Enhanced Prompt (CRITICAL)

Replace the current prompt in `~/.kiro/agents/anki-agent.json`. Key changes:

1. Fix `bible-tools` error → `koine-pimsleur/src/enrich.py` via `KOINE_ANKI_PATH`
2. Fix `Wavenet-B` error → Chirp3-HD fallback pool, char-count trigger
3. Add Architecture section (compounds pipeline, terminaciones formats, pimsleur deck selection)
4. Add 7 Critical Rules (NFC, stable IDs, registry dedup, vocabulary blacklist, TTS fallback, Spanish-first, `KOINE_ANKI_PATH`)
5. Add Build & Run with exact commands
6. Mark `anki-main/` as frozen

```
# Backup current prompt before replacing
cp ~/.kiro/agents/anki-agent.json ~/.kiro/agents/anki-agent.json.bak.$(date +%Y%m%d)
```

Action 2 — Create .kiro/steering.md (HIGH)

```
mkdir -p ~/work/github/anki/.kiro
cat > ~/work/github/anki/.kiro/steering.md << 'EOF'
# anki Project — Always-On Context

## Vocabulary Blacklist (Modern Greek — BANNED)
νερό-ύδωρ | σπίτι-οἶκος | ψωμί-ἄρτος | πόρτα-θύρα
All Turkish/Italian loanwords (post-1453) are banned.

## Greek Text Rule
Always store Greek as NFC: unicodedata.normalize('NFC', text)

## Active Repos
koine-anki/ and koine-pimsleur/ are active. anki-main/ is FROZEN.
EOF
```

Action 3 — Fix lessons-learned-koine.md TTS Section (MEDIUM)

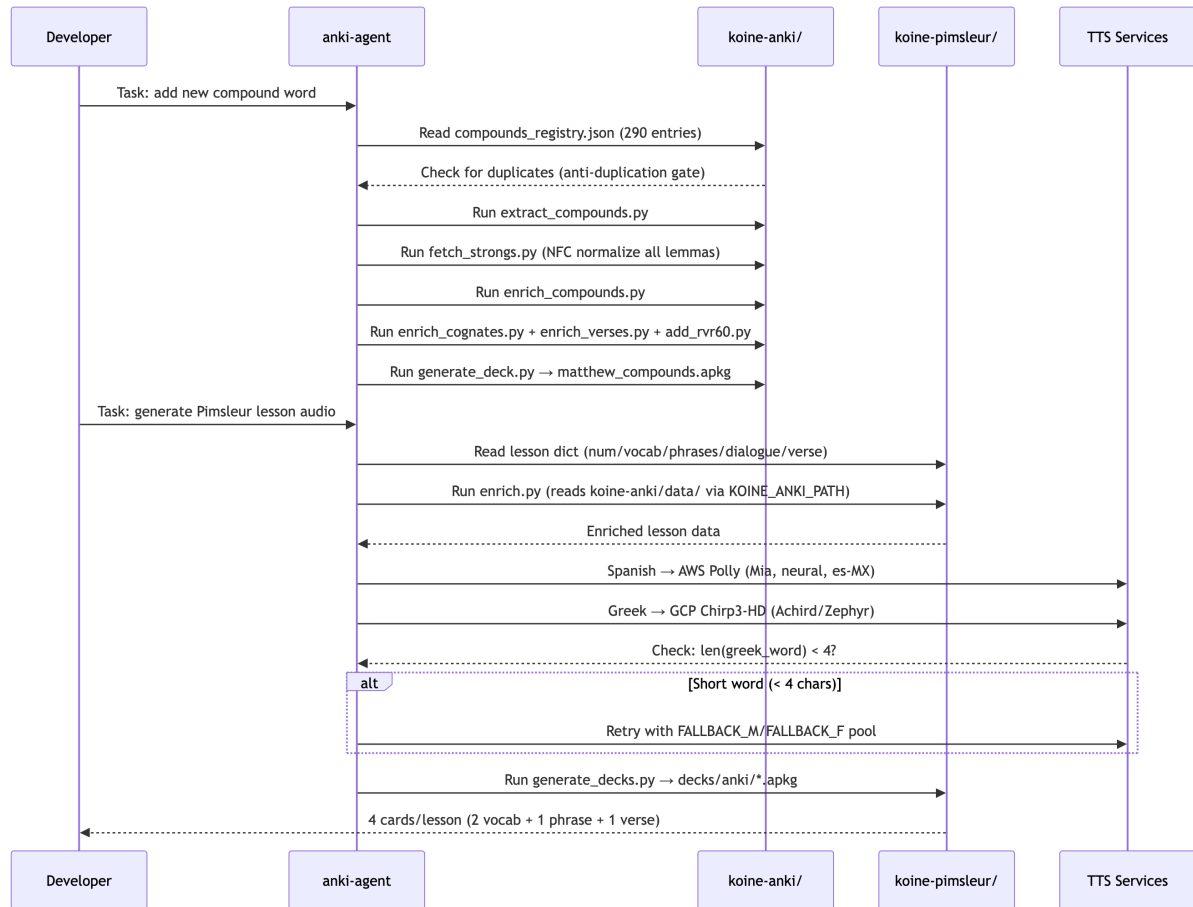
Update `anki-main/languages/notes/lessons-learned-koine.md`: - Remove “Wavenet-B fallback” claim
- Document actual behavior: char-count threshold (`< 4`), Chirp3-HD fallback pool - Document `_min_dur()`
formula: `max(400, min(n,4)*150 + max(0,n-4)*80)` ms

Action 4 — Document Lesson Data Format (LOW)

Create `koine-pimsleur/notes/lesson-data-format.md` documenting the lesson dict keys:

```
num, vocab, phrases, dialogue, verse, recon, closing_es
```

Prompt Enhancement Sequence



Summary Table

Finding	Severity	Status	Action
<code>bible-tools</code> reference in prompt	🔴 Error	Confirmed wrong	Fix in enhanced prompt
<code>Wavenet-B</code> fallback in prompt/notes	🔴 Error	Confirmed wrong	Fix prompt + notes
No <code>.kiro/steering.md</code>	🟡 Gap	Missing	Create steering file
4 new scripts not in prompt	🟡 Gap	Missing	Add to prompt
Lesson data format undocumented	🟡 Gap	Missing	Add to notes
Broken venv shebang in <code>anki-main</code>	🟡 Gap	Known	Use <code>python3 -m pip</code>
	✅ OK	Confirmed	Document in steering

Finding	Severity	Status	Action
NFC normalization — verified correct			
Stable deck IDs — verified correct	✅ OK	Confirmed	Never change
KOINE_ANKI_PATH dependency	✅ OK	Confirmed	Document in prompt

References

Source	Finding
<code>~/.kiro/agents/anki-agent.json</code>	Current prompt (1,159 chars, 2 errors)
<code>koine-pimsleur/src/enrich.py</code>	Actual consumer of <code>koine-anki/data/</code> via <code>KOINE_ANKI_PATH</code>
<code>koine-pimsleur/src/tts.py</code>	Actual TTS fallback: Chirp3-HD pool, char-count <code>< 4</code> trigger
<code>koine-anki/compounds/fetch_strongs.py</code>	NFC normalization at ingest
<code>koine-anki/compounds/generate_deck.py</code>	Compounds deck (model ID <code>1646410861305</code>)
<code>koine-anki/terminaciones/gen_deck.py</code>	TSV output (not .apkg)
<code>koine-anki/terminaciones/gen_deck_vol3.py</code>	.apkg output (model ID <code>1607392319</code>)
<code>koine-pimsleur/src/generate_decks.py</code>	Pimsleur deck (model ID <code>1646410900000</code>)
<code>koine-anki/data/compounds_registry.json</code>	290 Matthew entries, anti-duplication gate
<code>koine-anki/notes/card-quality-standards.md</code>	Card quality rules
<code>koine-anki/notes/koine-greek-anki-guidelines.md</code>	Spanish-first, simplified terminology
<code>anki-main/languages/notes/lessons-learned-koine.md</code>	Vocabulary blacklist (contains stale TTS info)